

PhiShark2026: A Multi-Layer Active-Web Raw-Evidence Dataset for Phishing Website Research

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Abstract—Phishing websites are short-lived and rapidly changing, yet many phishing datasets reduce observations to URLs or precomputed features, constraining researchers to predefined representations and discarding the underlying evidence needed to derive alternative features, apply new extraction methods, examine cross-layer relationships, and reanalyze observations as phishing techniques evolve. This study addresses this limitation with a multi-layer active-web dataset comprising 67,502 scans, including 33,387 phishing observations from operational feeds and 34,115 screened benign reference observations. The corpus preserves raw evidence across HTML content and screenshots, URL and redirect behavior, HTTP and security headers, compliance files, TLS certificates, DNS and domain registration, open ports, geolocation and accessibility measurements, and network infrastructure, while explicitly recording unavailable evidence rather than treating it as negative observations. To avoid misleading infrastructure attribution on shared platforms, the study applies a hosting-aware evidence model that masks provider-owned infrastructure signals for free-hosted tenant pages while retaining meaningful page- and transport-level evidence. Characterization reveals systematic differences between phishing and benign websites across web-resource usage, domain maturity, mail and policy configuration, security headers, and infrastructure context. By preserving raw artifacts together with acquisition metadata and explicit evidence availability, the corpus provides an inspectable and reproducible foundation for future phishing measurement and dataset research.

I. INTRODUCTION

Phishing remains one of the most persistent and financially damaging threats in the modern cybersecurity landscape. Recent industry intelligence indicates that a substantial share of successful cyberattacks, particularly those leading to severe data breaches and ransomware deployments, originate from phishing and related social engineering activity [1], [2]. The financial consequences of such incidents also remain high, with phishing-related breaches continuing to impose significant organizational costs [3]. These observations indicate that phishing remains a widespread threat affecting both individuals and organizations, despite sustained defensive efforts from academia and industry.

A central challenge lies in the mismatch between academic datasets and real-world operational conditions. Static datasets often do not adequately represent the dynamic evasion tactics, class imbalance, and structural variability of real-world phish-

ing campaigns [4]. The mismatch motivates closer examination of the datasets that continue to underpin phishing research.

A widely acknowledged limitation is that many benchmark datasets, including the UCI Phishing Dataset and standard Mendeley repositories, are released as pre-aggregated tables rather than preserved source artifacts [5], [6]. Such formatting restricts reproducibility and makes it difficult to revisit derived fields when phishing tactics evolve. Prior phishing-detection studies continue to rely heavily on URL-based lexical fields, often supplemented with basic HTML content and coarse SSL/TLS indicators [7], [8]. Although such intra-URL fields have historically yielded useful localized results, they are increasingly insufficient for capturing the sophistication of modern phishing [9], which now frequently involves techniques such as HTML smuggling [10] and decentralized hosting [11].

Phishing-website research therefore requires datasets that preserve evidence across multiple layers of the web ecosystem. Open-source intelligence (OSINT) techniques make it possible to collect contextual observations associated with URLs, domains, and IP addresses [12]. Such observations include DNS configuration, TLS certificate metadata, hosting infrastructure, network exposure such as open ports, and web-level artifacts. Examining these signals jointly supports measurement of how page-level, domain-level, and infrastructure-level properties interact without treating any individual field as a definitive detection signal.

This work addresses these gaps with broad multi-layer coverage over URL, web, certificate, and infrastructure evidence. Unlike commonly used static datasets, the corpus provides OSINT-derived data collected from URL, domain, and IP sources without restrictive prior aggregation. The preserved evidence includes DNS records, WHOIS and IP WHOIS data, TLS certificate metadata, open-port observations, HTTP headers, compliance-file analysis, webpage screenshots, raw HTML content, favicon data, redirection chains, cookies, and page statistics.

The dataset description is accompanied by a focused narrative synthesis of phishing datasets, evidence layers, and evaluation practice. The review positions the corpus against representative public benchmarks without claiming to constitute a new systematic review.

The paper contributes: (1) a 67,502-scan multi-layer raw-

evidence corpus intended for controlled research access; (2) a hosting-aware method separating tenant and provider evidence; (3) measurements across web, registration, TLS, security, and infrastructure layers; and (4) robustness analyses covering provider composition, domain duplication, page complexity, and threshold sensitivity, together with a cross-layer artifact-reuse analysis.

The remainder is organized as follows. Section II presents background, related datasets, measurement gaps, and design requirements. Section III describes the collection, from data sources and acquisition to artifact preservation and controlled access. Section IV characterizes the corpus evidence, and Section V isolates provider, duplication, complexity, and threshold effects. Section VI discusses implications and limitations, and Section VII concludes the paper, followed by the unnumbered Ethics Considerations section.

II. BACKGROUND AND RELATED DATASETS

Recent phishing-dataset research is methodologically active but empirically uneven. Much of the literature rests on datasets that are narrow in evidence depth, opaque in construction, or already transformed into fixed tables before release, without an equally strong basis for reproducibility, cross-layer interpretation, or operational realism.

A. Existing Phishing Datasets

The dataset landscape reinforces this concern. Table I summarizes representative phishing datasets introduced between 2014 and 2026 and contrasts them with the proposed controlled-access corpus.

The public benchmark tradition is broad, but it is not homogeneous. Older and still frequently reused resources such as the UCI Phishing Website dataset [14], ISCX-URL2016 [15], PWD2016 and related screenshot-aware releases [16], and Phish-IRIS [18] established baseline comparability early. More recent public releases such as PILWD-134K [19], LNU-Phish [22], PhishIntention [24], PhiUSIIL [27], MTLP [28], PhreshPhish [29], and Phish360 [31] expanded that coverage further. Taken together, these datasets show steady progress, but most public releases still emphasize URL, HTML, or simplified SSL fields more than transparent raw multi-layer evidence preservation.

The evolution can be read in three stages. The early public benchmark era is dominated by URL-centric or lightly extended tabular datasets, often combining lexical attributes with coarse SSL, HTML, or WHOIS fields [14], [15], [20]. These resources typically represented phishing as a fixed table rather than as an observable active-web phenomenon, making it difficult to revisit how the evidence had been acquired or encoded.

The second stage broadens the visible surface of the problem. Datasets such as PWD2016-type releases, PILWD-134K, LNU-Phish, and PhishIntention begin to incorporate screenshots or richer HTML context alongside URL-level attributes [16], [19], [22], [24]. This is an important step because phishing is not purely a suspicious-string problem;

it is also a rendered-page impersonation problem. Even so, these expansions are usually partial. Screenshot-aware datasets still do not always preserve the broader surrounding evidence needed to study registration context, infrastructure, certificate behavior, redirection logic, and rendered-page semantics together.

The third stage pushes toward larger or more realistic collections. Recent resources such as PhiUSIIL [27], MTLP [28], PhreshPhish, PhishXtract-Class, and Phish360 [29]–[31] improve scale, freshness, or modality coverage in different ways. Some move toward infrastructure-aware or screenshot-aware representations; some emphasize more contemporary active-web collection. However, even this newer generation usually stops short of releasing a reproducible, artifact-complete evidence stack that exposes how webpage content, visual-semantic behavior, registration patterns, certificate context, DNS structure, and hosting ecology interact.

B. Measurement Gaps

The central limitation in the literature is that the observable evidence base remains incomplete. Three recurring patterns stand out. First, many widely reused datasets are released in pre-processed tabular form rather than as artifact-complete collections. This is especially visible in UCI-style tables, Vrbancic’s dataset [20], PhiUSIIL [27], and related mirrors, which offer limited support for auditability or re-interpretation. Second, SSL information is often reduced to HTTPS presence or a few coarse certificate proxies, even though modern phishing pages routinely use valid and often free certificates. Third, screenshots appear in only a subset of public datasets, including PWD2016-type releases [16], PILWD-134K [19], LNU-Phish [22], PhishIntention [24], and Phish360 [31]. Even there, they are not usually paired with the broader evidence stack needed to study visual deception jointly with registration, hosting, and client-side behavior.

The second persistent limitation is partial modality coverage. A large share of phishing research still relies primarily on lexical URL patterns, basic HTML cues, or coarse SSL fields [7], [8], [32], [33]. These signals describe only a narrow slice of the phishing problem. They say little about how a page behaves when rendered, how its infrastructure is configured, how its certificate profile interacts with domain age, or whether the visual and semantic identity of the page is coherent with the surrounding web context.

The third limitation is that infrastructure-aware work is usually partial rather than holistic. Some studies do move beyond purely lexical analysis and incorporate DNS, WHOIS, TLS, IP, or related hosting signals. Work such as Fresh-Phish [34], LNU-Phish [22], domain-registration studies [35], traffic- or TLS-oriented representations [36], and domain-ownership analysis [30] demonstrates clear movement toward broader evidence families. Yet in many cases these fields are appended to an otherwise conventional table rather than released as raw, cross-layer artifacts that support alternative extraction logic, future re-analysis, or analyst-facing interpretation.

TABLE I
COMPARISON WITH REPRESENTATIVE PHISHING RESEARCH DATASETS

Dataset	Study	Year	Size	Data Source	Preserved artifacts
PhishStorm	[13]	2014	96K	URL, SSL	✓
UCI Phishing Website	[14]	2015	11K	URL, SSL, HTML, WHOIS	×
ISCX-URL2016	[15]	2016	45K	URL	✓
PWD2016	[16]	2016	30K	URL, SSL, HTML, Screenshot	✓
Ebbu 2017	[17]	2017	74K	URL	✓
Phishing Dataset	[16]	2018	30K	URL, SSL, HTML, WHOIS, Screenshot	✓
Phish-IRIS	[18]	2019	2.9K	Screenshot	✓
PILWD-134K	[19]	2020	134K	URL, SSL, HTML, Screenshot	✓
Vrbancic's Dataset	[20]	2020	89K	URL, SSL, HTML, WHOIS	×
Aljofey's Dataset	[8]	2021	60K	URL, SSL, HTML	✓
Phishing Websites	[21]	2021	80K	URL	✓
LNU-Phish	[22]	2022	~23K	URL, SSL, HTML, DNS, WHOIS, Screenshot	✓
VanNL126k	[23]	2021	~126K	URL, SSL, HTML, Screenshot	✓
PhishIntention	[24]	2023	~58K	URL, SSL, HTML, Screenshot	✓
PhiKitA-500	[25]	2023	2K	URL, SSL, HTML	✓
Tamal's Dataset	[26]	2024	248K	URL	×
PhiUSIIL Phishing URL	[27]	2024	236K	URL, SSL, HTML	×
MTLP Dataset	[28]	2024	100K	URL, SSL, HTML, WHOIS	✓
PhreshPhish	[29]	2025	~498K	URL, SSL, HTML	✓
PhishXtract-Class	[30]	2026	~5K	URL, SSL, DNS	×
Phish360	[31]	2026	~11K	URL, SSL, HTML, Screenshot	✓
PhiShark2026		2026	67,502	URL, TLS, HTML, Screenshot, Favicon, DNS, WHOIS/RDAP, Ports, IP/ASN, Redirects, HTTP Transactions, Security Headers, Compliance Paths	✓

Preserved artifacts indicates that a research release contains at least one source-level artifact, such as URLs, HTML, or screenshots, rather than only a precomputed feature table. A checkmark does not imply equal artifact breadth across datasets or unrestricted public access.

The fourth limitation is operational realism. Modern phishing increasingly combines convincing brand mimicry, transient hosting, rapidly rotated domains, free TLS certificates, templated landing pages, and short operational lifetimes [10], [11], [29]. Under these conditions, single-layer evidence can become brittle: a benign site may expose isolated suspicious-looking properties, while a phishing page may appear superficially clean on any one layer. Current public datasets have improved substantially, but they still rarely model active website detection as an evidence ecology spanning lexical, structural, rendered-page visual-semantic, certificate, registration, and infrastructure layers at the same time.

C. Design Requirements

Taken together, the recent literature suggests that a central bottleneck in phishing research is the shortage of well-documented, inspection-ready evidence ecosystems. Few datasets preserve enough context to examine which evidence layers are present, how observations interact, and whether apparent patterns remain meaningful after obvious shortcuts are controlled.

Against that background, the corpus is designed to address the gap directly. It preserves URL, HTML, screenshot, favicon, DNS, WHOIS, IP WHOIS, certificate, redirection, header, open-port, and compliance-related evidence within an active-site benchmark setting, which better matches the point at which many real phishing pages become analyzable.

The present study therefore does more than add columns to earlier datasets. It provides a benchmark in which raw evidence preservation, ecological interpretation, and free-host control can be studied together rather than in isolation, creating

a more transparent and ecologically grounded substrate for future phishing research.

III. PHISHARK2026

This collection is an active-web phishing corpus intended for controlled research access and designed around artifact preservation rather than a fixed derived table. Its unit of observation is a scan: a target URL acquired at a specific point in time together with the page, transport, registration, and infrastructure evidence successfully observed during that acquisition. This design allows approved researchers to audit collection failures, construct alternative representations, and revisit analytical assumptions, consistent with calls for transparent dataset documentation and more complete phishing-resource collection [37], [38].

A. Data Sources

Phishing targets are drawn from two operational sources: OpenPhish and the Malicious Links List, maintained by the Cybersecurity Presidency of the Republic of Türkiye [39], [40]. These feeds provide operationally reported targets rather than a complete census of phishing activity; source-specific collection intervals are reported with the dataset composition in Section III-F.

B. Acquisition Architecture

Each target is canonicalized into URL-, host-, and registrable-domain forms before a distributed acquisition pipeline dispatches browser, network, DNS, TLS, WHOIS/RDAP, geolocation, compliance-path, and port-observation workers. Browser and network tasks run through

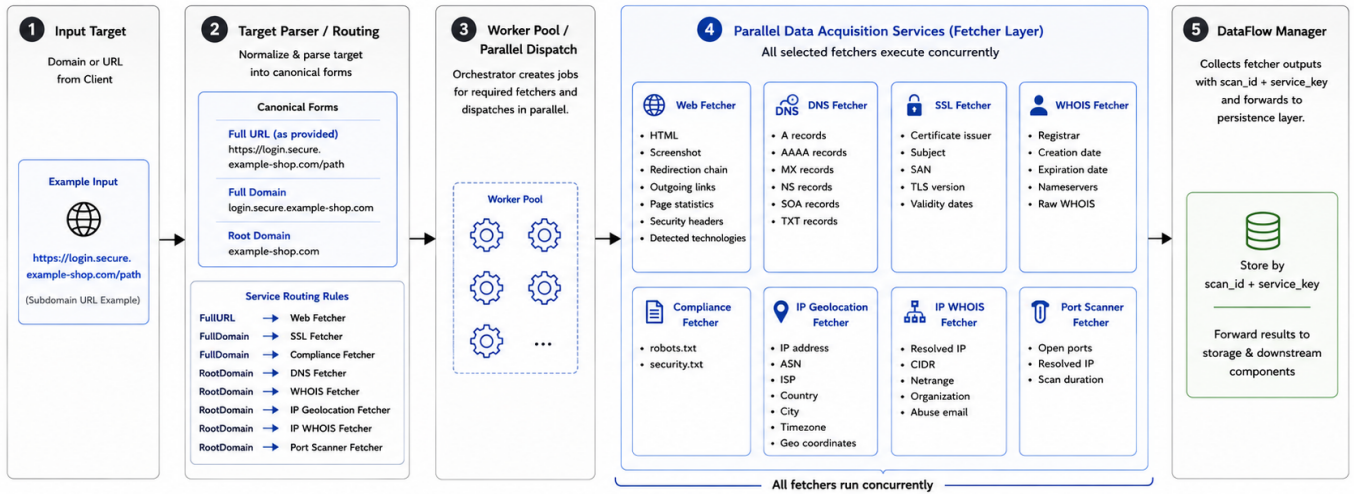


Fig. 1. Distributed active-web acquisition architecture used to build the corpus. Submitted targets are canonicalized, routed to parallel specialized fetchers, and persisted under a shared scan identifier.

TABLE II
FETCHER IMPLEMENTATIONS AND PRESERVED EVIDENCE.

Fetcher	Input and mechanism	Preserved evidence
Web	Full URL; headless Chromium through chromedp and the Chrome DevTools Protocol [41]	Rendered HTML, screenshot, redirects, outgoing links, detected technologies, and browser/network statistics
DNS	Registrable domain; miekg/dns with the Go resolver as fallback [42]	A, AAAA, MX, NS, TXT, CNAME, and SOA records
TLS	Full host; Go crypto/tls and crypto/x509	Issuer, subject, validity, serial number, SANs, and negotiated TLS version
Domain registration	Registrable domain; RDAP client and registry bootstrap [43]	Registrar, registration dates, nameservers, registrant country, DNSSEC, and raw response
Compliance	Full host; bounded HTTP retrieval of public well-known paths	Locations of security.txt, robots.txt, sitemap.xml, and related public files [44]
IP geolocation	Resolved IP; IP-API JSON endpoint [45]	Country, region, city, coordinates, timezone, ISP, organization, and AS
IP WHOIS	Resolved IP; WHOIS client and parser [46]	Network range, CIDR, organization, country, ASN/origin, abuse contact, and registration dates
Port observation	Resolved IPv4; ProjectDiscovery Naabu top-100-port profile [47]	Resolved address, observed open ports, and scan duration

a consistent collection policy with proxy-backed retrieval to reduce region-specific blocking and transient crawl gaps. All outputs are joined by a shared scan identifier. Figure 1 summarizes this routing and acquisition path.

The acquisition pipeline routes the canonical full URL, full host, or registrable domain to eight specialized fetchers. Table II records the acquisition mechanism and retained evidence for each fetcher.

C. Benign Reference Construction

The benign candidate pool was sampled at random from the OpenPageRank Top 10 Million Domains list, whose domain-level authority ranking is derived from the Common Crawl

web graph [48]. Independently registered candidates were selected by unrestricted random sampling from this pool. Free-hosted candidates were also selected randomly, but within provider strata chosen to approximate, as closely as feasible, the provider distribution observed in the phishing stratum. This matching preserves ordinary tenant pages on shared platforms while reducing provider-composition imbalance between the phishing and benign free-hosting groups.

Before inclusion, every benign candidate was screened independently with the Google Web Risk URL Lookup service [49] and the PhiShark URL-analysis platform [50]. Only candidates for which neither service returned a malicious or phishing indication were retained. This dual screen reduces obvious known-threat contamination but does not establish a perfect trust universe: newly deployed, uncatalogued, or misclassified pages can still survive both checks. The resulting pool is therefore treated as a curated reference class rather than ground-truth proof of benignness.

The PhiShark URL-analysis platform is an author-operated auxiliary service used for candidate screening. Its scores and derived outputs are not included in the released evidence and are not used in the reported characterization analyses.

Benign observations were assembled as targeted source-specific snapshots rather than a continuously sampled feed. They are therefore consolidated in the source accounting, and calendar time is not used for a two-class temporal comparison against the repeatedly collected phishing feeds.

D. Raw Artifact Preservation

The corpus preserves the original and normalized target URL; HTTP outcomes, headers, cookies, final landing targets, and redirect chains; raw HTML and extracted text; rendered screenshots and favicons; DNS records; WHOIS/RDAP and IP WHOIS outputs; TLS certificate metadata; nameserver and mail configuration; contacted web resources; lightweight open-port observations; regional accessibility outcomes; and public compliance-path results. Screenshot capture is required for

TABLE III
SOURCE PROVENANCE, COLLECTION INTERVALS, AND COMPLETE CLASS-BY-HOSTING DISTRIBUTION.

Source or summary	Class	Free host	Non-free	Total	Collection interval (UTC)
OpenPhish [39]	Phishing	14,633	16,245	30,878	2026-04-23 to 2026-08-15 (113.9 days)
Malicious Links List (Cybersecurity Presidency) [40]	Phishing	453	2,056	2,509	2026-06-14 to 2026-08-15 (62.1 days)
OpenPageRank [48]	Benign	15,359	18,756	34,115	Targeted snapshots
<i>Class and corpus totals</i>					
Phishing total	Phishing	15,086	18,301	33,387	
Benign total	Benign	15,359	18,756	34,115	
Overall total	All	30,445	37,057	67,502	

inclusion, whereas other layers retain their observed missingness.

Exact hashes are retained for transparency and reuse analysis, but they are not campaign labels.

E. Hosting-Aware Evidence Model

Free-hosted tenant pages and independently registered sites expose fundamentally different ownership boundaries. URL structure, HTML, screenshots, favicons, web retrieval, security headers, compliance paths, and TLS context remain meaningful page- or transport-level observations in both hosting strata. In contrast, DNS, WHOIS/RDAP, IP/ASN, server country, port exposure, and geo-accessibility often describe the provider when a page is hosted under a shared platform root. Prior large-scale measurements show that free website-building services are actively abused for phishing and differ operationally from self-hosted sites [29], [51]. These infrastructure layers are therefore compared only in the non-free-hosting stratum.

The evidence model uses layer-specific available-case denominators. Missing fetches remain missing and are not silently recoded as negative observations. This mask prevents shared platform characteristics from being attributed to tenant pages while retaining free hosting as an operationally important part of the dataset. Table IV reports the resulting scope and coverage.

F. Dataset Composition and Access

Table III combines source provenance, collection intervals, and the complete class-by-hosting distribution. OpenPhish contributes 30,878 scans collected between April 23 and August 15, 2026, while the Malicious Links List contributes 2,509 scans collected between June 14 and August 15, 2026. All benign reference candidates were randomly sampled from OpenPageRank; free-host selections additionally followed the provider-stratified design described above. The final corpus contains 33,387 phishing and 34,115 benign scans; 30,445 use free hosting and 37,057 use non-free hosting.

Table IV makes the analytical denominator explicit for each layer. The lower WHOIS/RDAP and infrastructure coverage on phishing scans is itself a collection property and is kept separate from substantive class comparisons.

Release organization and record linkage. The controlled package contains a release metadata CSV and one ZIP archive per retained scan. The CSV uses `scan_id` as a unique

primary key and records the corresponding relative archive path, class and source provenance, hosting stratum, target and final-page identifiers, scan timestamps and status, and artifact-presence flags. For all 67,502 retained records, the corresponding archive is named `<scan_id>.zip`. Table V summarizes how that identifier connects the release components.

Approved researchers can first filter the release metadata CSV and obtain the unique `scan_id` of each selected record, then open the corresponding `<scan_id>.zip` archive. Within the archive, `metadata.json` records scan identity and artifact references, `data/*.json` contains the available fetcher-specific observations, and `artifacts/` contains the captured screenshot, favicon, and losslessly compressed HTML. Evidence extracted from different modules is joined using the enclosing scan identifier. An absent optional module or artifact is treated as unavailable evidence rather than as a negative observation.

Dataset documentation and its access-request interface will be publicly available through <https://phishark2026dataset.com> [52], while the raw corpus will be distributed only to approved research users. The controlled research package covers the preserved raw evidence described above. Its release documentation will describe the metadata schema, per-scan archive layout, evidence modules, and access instructions so that approved users can inspect and analyze the provided artifacts independently.

IV. DATASET CHARACTERIZATION

This section characterizes the corpus using normalized raw-fetcher observations, release metadata, and captured artifacts. To keep the evidence proportional to the claim, tables report exact comparisons, while figures are reserved for distributions, reversals, or uncertainty that are difficult to read from individual cells.

The analysis follows the collection boundary established in Section III. Page-level observations are compared within both hosting strata. Provider-owned infrastructure (DNS, WHOIS/RDAP, IP geolocation, ports, and regional accessibility) is analyzed only for independently registered, non-free-hosting sites. Each result uses its layer-specific available-case denominator; a missing fetch is never recorded as an observed negative.

The characterization proceeds from directly observed page structure to infrastructure context and exact artifact reuse.

TABLE IV
EVIDENCE AVAILABILITY AND HOSTING-AWARE ANALYTICAL SCOPE. PERCENTAGES USE THE CLASS-LEVEL SCAN DENOMINATOR; SUBSTANTIVE COMPARISONS USE THE LAYER-SPECIFIC AVAILABLE POPULATION.

Evidence layer	Analysis scope	Phishing availability	Benign availability	Treatment
Screenshot	All strata	33,387/33,387 (100.0%)	34,115/34,115 (100.0%)	Required inclusion artifact
HTML	All strata	33,099/33,387 (99.1%)	33,795/34,115 (99.1%)	Page-level; available-case denominator
Favicon	All strata	23,619/33,387 (70.7%)	24,539/34,115 (71.9%)	Optional artifact; missing is not negative
Web fetch	All strata	33,387/33,387 (100.0%)	34,115/34,115 (100.0%)	Page-level retrieval comparison
TLS	All strata	31,853/33,387 (95.4%)	33,639/34,115 (98.6%)	Transport context; hosting-stratified
Security headers	All strata	33,387/33,387 (100.0%)	34,115/34,115 (100.0%)	Hosting-stratified posture
Compliance paths	All strata	32,846/33,387 (98.4%)	33,719/34,115 (98.8%)	Secondary; bodies not retained
DNS	Non-free only	18,089/18,301 (98.8%)	18,756/18,756 (100.0%)	Shared free-host infrastructure masked
WHOIS/RDAP	Non-free only	14,329/18,301 (78.3%)	17,583/18,756 (93.7%)	Complete-case registration analysis
IP geolocation	Non-free only	14,795/18,301 (80.8%)	17,905/18,756 (95.5%)	Observed hosting context only
Port scan	Non-free only	15,651/18,301 (85.5%)	17,905/18,756 (95.5%)	Observed host exposure only
Geo accessibility	Non-free only	18,301/18,301 (100.0%)	18,756/18,756 (100.0%)	Six-region comparison

TABLE V
CONTROLLED RELEASE ORGANIZATION AND RECORD LINKAGE

Release item	Linkage	Role
Metadata CSV	scan_id	Record index, archive path, provenance, hosting stratum, scan status, and artifact-presence flags
<scan_id>.zip	ZIP filename	One preserved active-web scan
metadata.json	Archive context	Scan identity, schema, and artifact paths and hashes
data/*.json	Archive context	Fetcher-specific observational evidence
artifacts/*	Metadata paths	Screenshot, favicon, and losslessly compressed HTML

Within each subsection, the analytical comparison is stated first, the principal result is interpreted second, and the supporting table or figure follows. Section V then evaluates composition sensitivity separately. Results are descriptive unless an effect estimate and uncertainty interval are explicitly reported.

A. Web-Resource Ecology

Page footprint and retrieval.

Analysis design. We compare page-shape and crawler-observed navigation fields within the four hosting-by-class strata. Medians summarize skewed size and count variables; rates are used for binary retrieval outcomes. These measurements describe the captured page, not malicious intent.

Result analysis. Phishing target URLs are longer at the median in both strata: 43 versus 30 characters in free hosting and 37 versus 27 in non-free hosting. Non-free phishing pages are markedly lighter than benign pages: median HTML payload is 30.4 versus 78.8 KB, screenshot size is 130.2 versus 270.4 KB, and extracted URL count is 18 versus 71. Free-hosting page weight behaves differently because tenant pages inherit platform templates. Redirects are widespread in every group, while final-host changes are less frequent for phishing, particularly in non-free hosting. Table VI reports the complete main-paper comparison.

Non-free web-resource ecology.

Analysis design. We next examine the browser transaction graph for non-free-hosting scans. The fields count requests, scripts, cookies, contacted domains, IPs, ASNs, and transferred bytes. Exact medians are more informative here than a set of near-identical bars, so Table VII carries the comparison.

Result analysis. The resource graph is consistently thinner for phishing. Median phishing pages contain 16 HTTP trans-

actions, 3 script transactions, 1 cookie, 3 resource domains, and 181.3 KB of transferred resources, compared with 43, 11, 3, 5, and 691.8 KB for benign pages. HTTP error transactions are the exception: their median is one for phishing and zero for benign pages.

A descriptive minimal interaction stack, defined as at most 20 transactions, 3 scripts, 1 cookie, and 3 resource domains, occurs in 33.3% of non-free phishing scans and 12.9% of benign scans. The complementary rich stack occurs in 4.3% and 31.2%, respectively. These thresholds summarize a multimetric phenotype; they are neither optimized cutoffs nor deployment rules. Free hosting reverses several resource-complexity relationships, reinforcing the decision to preserve hosting strata.

Detected technologies.

Analysis design. Technology detections are first standardized within observed hosting provider and then repeated after registration-domain deduplication. This separates broad operational integrations from repeated-domain artifacts.

Result analysis. Non-free phishing scans contain fewer detected technologies overall: median 1 and mean 2.03, compared with median 3 and mean 3.04 for benign scans. Provider-standardized prevalence remains benign-high for tag managers (9.62% versus 49.41%), analytics (28.33% versus 56.76%), JavaScript libraries (35.91% versus 56.42%), and content-management systems (5.70% versus 27.92%). Figure 2 is retained because it reveals a result that a point-estimate table obscures: Angular, Ruby on Rails, and React are phishing-high at scan level by 8.76, 8.66, and 7.23 percentage points, but reverse to -0.99 , -1.67 , and -2.21 points after registration-domain deduplication. These frameworks must not be described as intrinsically risky.

Technology sparsity partly overlaps the minimal interaction stack. Minimal phishing pages contain 0.73 detected technologies on average compared with 2.67 among non-minimal phishing pages; the benign means are 0.57 and 3.41. Within-class correlations between minimal-stack membership and technology count are -0.365 for phishing and -0.414 for benign. The two observations are therefore related rather than independent evidence.

TABLE VI

PAGE FOOTPRINT AND WEB RETRIEVAL BEHAVIOR. SIZE AND COUNT ENTRIES ARE MEDIANS; QUERY, REDIRECT, AND HOST-CHANGE ENTRIES ARE SCAN-LEVEL RATES.

Observation	Free phish.	Free benign	Non-free phish.	Non-free benign
Target URL length (characters)	43	30	37	27
HTML payload (KB)	12.5	11.4	30.4	78.8
Screenshot file size (KB)	169.1	74.3	130.2	270.4
Extracted URL count	12	19	18	71
Target URL contains query	4.5%	0.3%	8.7%	6.4%
Redirect observed	84.6%	99.4%	75.8%	87.7%
Final host changed	8.3%	9.1%	14.4%	23.3%

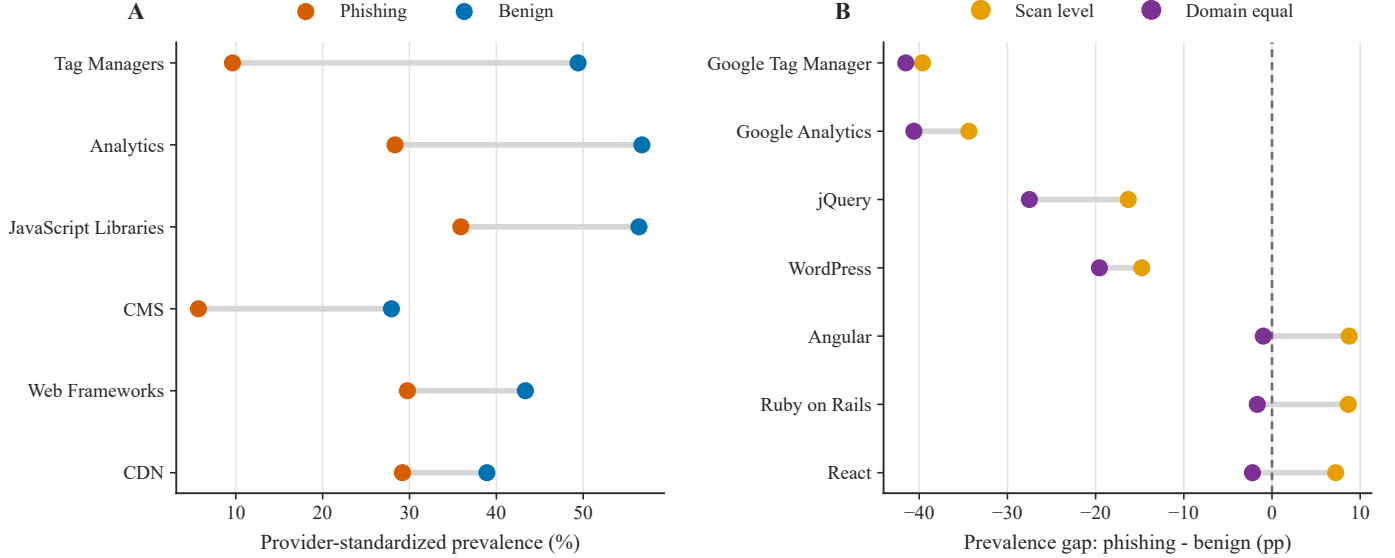


Fig. 2. Detected-technology ecology in non-free hosting. Left: provider-standardized category prevalence. Right: scan-level and registration-domain-equal prevalence gaps for selected technologies.

TABLE VII
MEDIAN NON-FREE-HOSTING WEB-RESOURCE ECOLOGY.

Observed metric	Phish.	Benign
HTTP transactions	16	43
Script transactions	3	11
Cookies	1	3
Resource domains	3	5
External resource domains	2	4
Resource IPs	3	6
Resource ASNs	2	3
Transferred resources (KB)	181.3	691.8
Observed page resources (KB)	167.6	626.7
HTTP error transactions	1	0

B. DNS and Registration

Analysis design. DNS comparisons use 18,089 phishing and 18,756 benign covered non-free scans. WHOIS/RDAP maturity uses available creation dates and is presented as a distribution rather than a single age threshold.

Result analysis. Nameserver and ordinary resolution records are common in both classes, but mail and policy posture differ. MX occurs in 29.7% of phishing and 86.7% of benign scans; TXT in 42.0% and 58.6%; and SPF in 28.4% and 50.7%. The mean number of populated record families is 3.82 versus 4.68. Thus, phishing domains are not simply “missing DNS”; they more often expose a thinner mail and policy profile. Table VIII

TABLE VIII
DNS ECOLOGY AMONG COVERED NON-FREE SCANS.

Observation	Phishing	Benign
A present	86.5%	95.5%
AAAA present	23.1%	29.4%
MX present	29.7%	86.7%
TXT present	42.0%	58.6%
NS present	97.9%	98.9%
SPF present	28.4%	50.7%
Mean populated families	3.82	4.68

gives the exact comparison.

Among DNS-covered scans, 2,435 phishing and 849 benign records have no A answer. None has a CNAME answer, and only one phishing and three benign cases have AAAA. CNAME-only or IPv6-only service therefore does not explain the no-A cases in this release; transient resolution state, incomplete resolver output, proxy behavior, and collection timing remain plausible.

WHOIS/RDAP is available for 14,329 phishing and 17,583 benign non-free scans; usable creation dates cover 13,785 and 14,981. Figure 3 follows the maturity result rather than preceding it. Phishing domains are distributed across young and mid-age buckets, with 21.0% at 0–7 days, whereas 91.5% of benign domains are older than five years. These are maturity

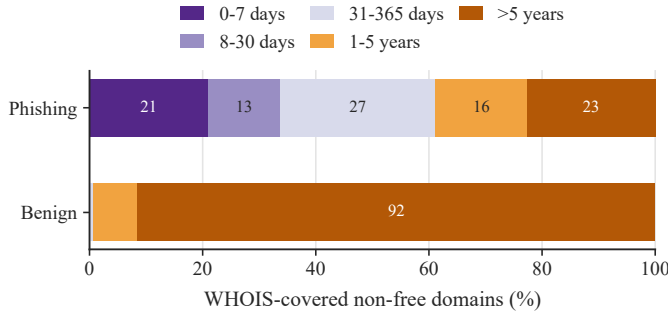


Fig. 3. WHOIS/RDAP-derived domain-age distribution for covered non-free-hosting scans.

TABLE IX
TOP OBSERVED PORTS IN COVERED NON-FREE-HOSTING SCANS (15,651 PHISHING; 17,905 BENIGN).

Port	Service	Phish.	Benign	Gap (pp)
80	HTTP	96.3%	97.8%	-1.5
443	HTTPS	94.8%	97.9%	-3.1
8443	HTTPS alt.	37.0%	30.6%	+6.4
8080	HTTP alt.	30.2%	29.6%	+0.6
22	SSH	33.4%	13.4%	+20.0
21	FTP	25.0%	14.7%	+10.3
993	IMAPS	25.0%	11.5%	+13.5
995	POP3S	24.9%	11.2%	+13.7

TABLE X
TLS ISSUER AND VALIDITY-WINDOW CONTEXT AMONG TLS-COVERED SCANS.

TLS observation	Free P	Free B	Non-free P	Non-free B
Let's Encrypt issuer	47.4%	52.9%	68.0%	38.3%
Google Trust issuer	46.3%	35.5%	22.8%	21.4%
Validity ≤ 91 days	94.1%	88.6%	92.8%	61.3%
Validity > 180 days	5.8%	11.2%	6.9%	38.2%

patterns, not deterministic labels: young benign domains and long-lived compromised domains remain possible [35], [53].

C. TLS, Security, and Infrastructure

Result analysis. Ports 80 and 443 dominate both covered groups. Several mail and remote-service ports are more prevalent in phishing scans, and median open-port count is 4 for phishing versus 3 for benign. This is host-exposure context rather than a monotonic risk measure. Table IX reports the leading ports.

TLS is transport context rather than proof of ownership or legitimacy, because valid certificates are now common within the phishing ecosystem [54], [55]. Let's Encrypt accounts for 68.0% of covered non-free phishing scans and 38.3% of benign scans. The free-hosting issuer mix includes a large Google Trust component, consistent with platform-mediated issuance. Short validity windows dominate phishing TLS: 92.8% of non-free phishing certificates are valid for at most 91 days, while 38.2% of benign certificates exceed 180 days. Table X reports the compact comparison.

Security headers and observed hosting location.

Result analysis. In non-free hosting, CSP is observed in 15.0% of phishing and 39.8% of benign scans, HSTS in 31.4%

and 55.9%, and X-Content-Type-Options in 28.5% and 56.5%. The distribution in Figure 4 is retained because it shows the completeness shift across both hosting strata: zero supported header families occur in 44.3% of non-free phishing and 22.9% of benign scans, while six or more occur in 8.1% and 28.5%. Free-hosting behavior differs because platform defaults can reverse individual relationships.

CSP presence is not equivalent to a restrictive policy; policy directives determine the protection actually provided [56]. Among non-free scans with CSP, `unsafe-inline` appears in 77.2% of phishing and 62.2% of benign policies, while `object-src 'none'` appears in 17.6% and 40.4%. Compliance-path results are also secondary because response bodies were not retained: `robots.txt` is fetcher-positive in 54.8% of non-free phishing and 71.7% of benign scans, but catch-all routes and soft-404 responses cannot be excluded.

Geo accessibility and server country describe the observed hosting environment, not actor origin. Full six-region accessibility is common in both groups; partial accessibility is more frequent among phishing scans. Canada and the United States dominate observed server locations, while Hong Kong is disproportionately represented among phishing hosts. Table XI keeps these small categorical results together.

D. Cross-Layer Artifact Reuse

Analysis design. Exact SHA-256 reuse is first measured within each artifact layer. Higher-confidence candidates then require the same combined signature to recur under at least two registration domains. Prior phishing-kit studies likewise use fingerprints and family-level grouping to connect recurring artifacts [53]; the intersections here are therefore reported as candidates, not confirmed campaigns.

Result analysis. Among available non-free artifacts, 89.7% of phishing favicons and 50.6% of phishing screenshots belong to repeated exact-hash clusters, compared with 20.6% and 8.9% for benign artifacts. Exact compressed-HTML reuse is much lower: 3.9% for phishing and less than 0.1% for benign. Single hashes still collide on generic pages, blank renders, default favicons, shared providers, and certificate bundles, echoing known limitations of presentation-level similarity evidence in real-world phishing collections [57].

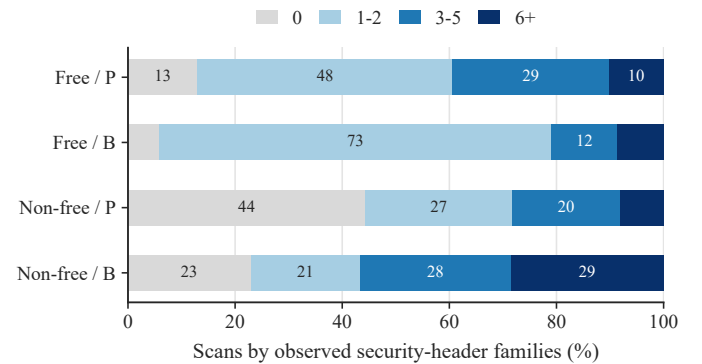


Fig. 4. Security-header completeness by hosting context and class. Counts cover twelve header families observed by the raw web fetcher.

TABLE XIII
SELECTED ROBUSTNESS RESULTS. AUC IS ORIENTED PHISHING-HIGH; VALUES BELOW 0.5 INDICATE LARGER OR MORE FREQUENT BENIGN OBSERVATIONS.

Check	Observation	Estimate	95% interval	Direction	Reading
Provider standardization	Minimal interaction stack	+21.96 pp; OR 3.80	N/A	Phishing-high	33.37% versus 11.41%; 14/15 providers agree
Provider-stratified AUC	HTTP transactions	0.290	0.198–0.323	Benign-high	Result is not driven by provider mix
Provider-stratified AUC	Script transactions	0.287	0.189–0.316	Benign-high	Thinner phishing script graph
Provider-stratified AUC	Transferred resources	0.323	0.224–0.355	Benign-high	Lower transferred volume for phishing
Provider-stratified AUC	HTTP error transactions	0.624	0.567–0.663	Phishing-high	Principal resource-ecology exception
Domain equal weighting	Core observations	22/22 retain direction	N/A	Mixed	Repeated domains change magnitude, not direction
Complexity-controlled AUC	Domain age	0.082	0.060–0.114	Benign-high	Strongest adjusted separation
Complexity-controlled AUC	DNS MX present	0.271	0.197–0.359	Benign-high	Mail posture persists among sparse pages
Complexity-controlled AUC	Transferred resources	0.387	0.330–0.428	Benign-high	Resource volume remains informative
Complexity-controlled AUC	Target URL length	0.674	0.602–0.764	Phishing-high	URL shape persists under adjustment
Complexity-controlled AUC	HTML size	0.612	0.536–0.685	Phishing-high	Direction reverses inside the sparse subset
Threshold sensitivity	Minimal-stack definitions	+12.81 to +33.37 pp	81 definitions	Phishing-high	Every tested threshold combination remains positive

TABLE XI
GEO ACCESSIBILITY AND LEADING OBSERVED SERVER-COUNTRY CODES FOR NON-FREE-HOSTING SCANS.

Layer	Category	Phishing	Benign
Reachability	All 6 regions	17,599 (96.2%)	18,688 (99.6%)
	3–5 regions	669 (3.66%)	63 (0.34%)
	0–2 regions	33 (0.18%)	5 (0.03%)
Server code	CA	3,769 (25.5%)	5,033 (28.1%)
	US	3,083 (20.8%)	7,056 (39.4%)
	HK	1,441 (9.7%)	20 (0.1%)
	NL	800 (5.4%)	399 (2.2%)
	DE	754 (5.1%)	1,087 (6.1%)

Table XII therefore reports multi-layer intersections with their eligible denominators. Screenshot-plus-favicon signatures cover 32.67% of eligible non-free phishing scans and 2.07% of benign scans. Screenshot-plus-hosting-IP covers 13.52% and 0.93%; favicon-plus-TLS-serial covers 9.49% and 0.57%. The strict screenshot-plus-favicon-plus-HTML signature identifies 46 phishing scans and no benign scans.

Multi-layer agreement sharply reduces benign collisions and yields stronger candidates for manual validation than any single hash. These groups remain candidates because shared infrastructure and generic artifacts can still create cross-domain collisions.

V. ROBUSTNESS ANALYSIS

The characterization results are descriptive properties of the collected evidence. This section tests whether their direction

TABLE XII
CROSS-DOMAIN EXACT-REUSE CANDIDATES IN NON-FREE HOSTING. SIGNATURES MUST RECUR UNDER AT LEAST TWO REGISTRATION DOMAINS.

Combined signature	Phishing	Benign
Screenshot + favicon	4,249/13,006 (32.67%)	337/16,286 (2.07%)
Screenshot + hosting IP	2,000/14,795 (13.52%)	166/17,905 (0.93%)
TLS certificate serial	2,053/17,434 (11.78%)	256/18,559 (1.38%)
Favicon + TLS serial	1,202/12,670 (9.49%)	93/16,259 (0.57%)
Screenshot + TLS serial	422/17,434 (2.42%)	53/18,559 (0.29%)
Screenshot + favicon + HTML	46/12,970 (0.36%)	0/16,235 (0.00%)

is explained by four important composition choices: hosting-provider mix, repeated registration domains, page complexity, or the exact threshold used to define a minimal interaction stack. Table XIII summarizes the principal estimates.

A. Provider Confounding

Provider standardization compares phishing and benign scans within observed hosting providers before aggregating the class difference. The minimal interaction stack remains more prevalent for phishing after this adjustment: standardized prevalence is 33.37% for phishing and 11.41% for benign, the Mantel-Haenszel odds ratio is 3.80, and 14 of 15 eligible providers retain the phishing-high direction.

The provider-stratified AUC results preserve the benign-high direction for HTTP transactions, script transactions, and transferred-resource volume. HTTP error transactions remain the principal exception and are phishing-high. The thinner resource ecology is therefore not explained only by different provider mixtures.

B. Domain Duplication

Registration-domain equal weighting gives each registrable domain the same aggregate influence, preventing repeatedly scanned domains from dominating the comparison. All 22 core observations retain their direction under this weighting, although several magnitudes change.

The technology comparison in Figure 2 illustrates why this control matters. Angular, Ruby on Rails, and React appear phishing-high at scan level but reverse after registration-domain equal weighting. These scan-level associations reflect repeated-domain composition and must not be interpreted as intrinsic framework risk.

C. Complexity Control

The harder complexity-controlled comparison restricts both classes to sparse pages and adjusts within provider and detected-technology-count strata. Domain age and MX presence remain strongly benign-high, and transferred-resource

volume remains benign-high. Target URL length and HTML size are phishing-high within the controlled sparse-page population; the HTML-size direction therefore differs from the pooled comparison.

Several other pooled differences, including non-web port exposure, HSTS, detected technology count, and Google Tag Manager, attenuate toward chance. Complexity control separates properties that persist among comparably sparse pages from signals that largely track overall page richness.

D. Threshold Sensitivity

The minimal interaction stack is re-evaluated across 81 combinations of transaction, script, cookie, and resource-domain thresholds. Every tested definition retains a phishing-high prevalence gap, ranging from +12.81 to +33.37 percentage points. The result is therefore not an artifact of the single descriptive threshold used in Section IV.

Temporal robustness is intentionally omitted. Phishing was collected repeatedly from operational feeds over approximately 114 days, whereas benign observations were assembled in short source-specific bursts. Short phishing lifetimes make capture timing consequential [38], [53], and a two-class temporal split here would confound calendar time with source and acquisition design.

VI. DISCUSSION AND LIMITATIONS

Interpretation.

The dataset contains 67,502 scans with near-balanced phishing and benign scan counts and, critically, comparable free-hosting and non-free-hosting strata. It therefore has a clear analytical boundary: page-level and transport observations can be studied across both hosting contexts, while provider-owned infrastructure is masked for free-hosted tenant pages.

The analysis adopts an evidence-ecology view rather than relying on isolated median comparisons. Non-free phishing pages are consistently thinner across HTTP transactions, scripts, cookies, contacted domains, IPs, ASNs, and transferred bytes. Provider stratification, registration-domain equal weighting, threshold sensitivity, and complexity control show which differences are stable and which are composition-sensitive. The technology analysis is especially instructive: operational integrations remain benign-high, while apparent Angular, Ruby on Rails, and React excesses reverse after registration-domain deduplication. This prevents repeated campaign sampling from being misreported as framework risk.

The infrastructure results are contextual rather than attributive. DNS and registration evidence show thinner mail/policy configuration and younger domains among phishing scans, but nameservers and ordinary resolution remain common. Port exposure, TLS issuer families, server-country codes, and geo accessibility describe the observed hosting environment; none identifies an attacker, operator, or country of origin. The same caution applies to exact artifact reuse. Multi-layer intersections provide stronger campaign candidates than a single screenshot or favicon hash, yet they remain candidates until generic pages, shared-provider defaults, and certificate bundles are excluded.

Dataset strengths.

The corpus preserves the evidence needed to audit and reinterpret the collection. Required screenshots, near-complete HTML, raw web transactions, TLS context, and layer-specific DNS, WHOIS/RDAP, IP, port, and geo observations support analyses that cannot be reconstructed from a fixed derived table. Explicit availability denominators prevent missing fetches from being treated as observed negatives, following established recommendations to document dataset composition, collection, and limitations [37].

The benign collection is deliberately heterogeneous and includes free-hosting pages. This is important because excluding benign tenant pages would make shared platforms appear class-specific; prior measurements show that these platforms are used by both ordinary publishers and phishing operators [51]. Likewise, the split between page-level and provider-level evidence prevents shared infrastructure from dominating class comparisons. Approved users receive the retained exact hashes and cross-domain reuse structure, enabling future campaign-oriented validation while documenting collision risks.

Limitations.

Active-web collection is temporally unstable. Pages can disappear, change content, geofence visitors, block crawlers, or return different artifacts by browser state and proxy location; short phishing lifetimes and collection failures are documented challenges for webpage datasets [38], [53]. OpenPhish and the Malicious Links List represent operational visibility and reporting practices rather than the complete phishing universe. The benign reference set is not a perfect trust universe and may contain reputational heterogeneity or residual label noise. Because the benign reference population originates from a popularity-ranked domain pool, registration-age and infrastructure comparisons may partially reflect reference-set composition rather than phishing behavior alone.

Layer availability is uneven. WHOIS/RDAP, IP geolocation, and port observations are less complete for phishing than for benign scans, so those analyses use available-case denominators and cannot assume that missingness is random. Compliance-path positives are not body-validated and may include catch-all routes or soft-404 responses. Detected technologies depend on observable scripts and page components, creating partial overlap with the resource-minimality result.

Repeated domains and artifacts remain part of the corpus. Registration-domain equal weighting preserves the direction of the core raw observations, but campaign- and template-cluster-aware subsets are still valuable for future reuse studies. Temporal class comparisons are deliberately omitted because repeated phishing-feed acquisition and burst-based benign acquisition have incompatible cadence. No claim of temporal stability, causal effect, deployment performance, or country/provider maliciousness is made.

Future dataset work.

Future controlled releases should extend the corpus longitudinally, preserve validated compliance response bodies, and provide explicit domain- and artifact-cluster groupings to approved users. Repeated captures of the same registrable

domains would make activation, dormancy, migration, and takedown observable without conflating class with collection schedule, as demonstrated by prior end-to-end phishing life-cycle measurements [53]. Manual validation of high-purity multi-layer reuse groups and content-aware similarity around near-duplicate artifacts would also strengthen campaign-level research beyond exact hashing.

VII. CONCLUSION

This collection provides 67,502 active-web scans with preserved evidence from URL, rendered page, HTTP, security-header, TLS, DNS, registration, hosting, port, geo-accessibility, and artifact layers. It contains 33,387 phishing and 34,115 benign observations and is organized into free-hosting and non-free-hosting strata so that shared provider infrastructure is not attributed to tenant pages.

Across the raw-evidence analyses, the dataset’s value lies in relationships among layers rather than in any single field. Non-free phishing pages have a consistently thinner web-resource ecology, younger registration profiles, and less complete mail, policy, and security-header posture. These directions survive provider adjustment and registration-domain equal weighting, while complexity control identifies which differences persist among equally sparse pages. The technology analysis demonstrates the importance of sensitivity testing: several scan-level framework associations reverse after domain deduplication and cannot be treated as intrinsic framework risk.

Artifact preservation also enables transparent reuse analysis. Exact screenshot and favicon reuse is common in phishing, but generic error pages and blank renders show why single hashes are not campaign labels. Cross-domain intersections such as screenshot-plus-favicon and screenshot-plus-hosting-IP substantially reduce benign collisions and provide stronger candidates for manual campaign validation.

The corpus remains bounded by operational-feed coverage, active-web volatility, incomplete layers, and incompatible phishing/benign acquisition cadence. Within those limits, the release offers approved researchers an inspectable and reproducible substrate for hosting-aware phishing research. Future controlled releases can build on the same raw evidence with longitudinal captures, validated public-file content, and domain- and artifact-cluster annotations.

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ETHICS CONSIDERATIONS

The corpus preserves raw web evidence, including full URLs, raw HTML, screenshots, favicons, redirection traces, and related page-level artifacts, because sanitizing these fields can remove information required for reproducibility and alternative analysis. It is therefore intended for controlled research access rather than unrestricted public download. Preserving

these artifacts improves auditability but creates clear dual-use risk.

Dual-use and residual risk.

Some retained artifacts may remain directly sensitive. Phishing URLs can contain victim-specific parameters, tokens, or campaign identifiers; raw HTML can contain embedded forms, hidden fields, attacker-controlled scripts, or operational kit structure; screenshots may capture personalized or contact-like content; and selected cookie, header, or redirection fields can reveal implementation patterns useful beyond defensive research. Controlled access reduces exposure but does not make the dataset risk-minimized.

Controlled research access.

Access is provided through a controlled request process rather than a direct download. Applicants provide their name, institutional affiliation, institutional email address, research purpose, and intended use. Requests are manually reviewed, and the raw corpus is distributed only to approved research users. Approval is based on a legitimate research purpose, identifiable affiliation, and agreement to the stated handling and non-redistribution conditions. No claim is made that every credential-harvesting trace, victim-linked parameter, or incidental personal datum has been removed.

Approved users must agree to use the corpus only for defensive or security research, handle it in isolated environments, refrain from replaying retained phishing artifacts or using them for credential collection or other offensive activity, and not redistribute the dataset to unauthorized third parties. Access may be revoked in cases of misuse. Publicly available documentation will provide safe-handling guidance and channels for privacy, abuse, and takedown requests, consistent with dataset-documentation practices that require intended uses, risks, and limitations to be made explicit [37].

The phishing portion is derived from operational feeds such as OpenPhish and the Malicious Links List, maintained by the Cybersecurity Presidency of the Republic of Türkiye [39], [40], so many samples already belong to an ecosystem of reporting and response. However, upstream feed inclusion is not treated as a full safety guarantee. The intended scientific benefit is narrower: the raw-evidence dataset supports transparent phishing research, but that benefit comes with residual privacy, misuse, and exposure risk that must be acknowledged explicitly.

REFERENCES

- [1] Cybersecurity and Infrastructure Security Agency (CISA), “Phishing guidance: Stopping the attack cycle at phase one,” Cybersecurity and Infrastructure Security Agency, Tech. Rep., 2024.
- [2] Verizon, “Data breach investigations report (dbir),” Verizon Enterprise Solutions, Tech. Rep., 2024.
- [3] IBM Security, “Cost of a data breach report,” IBM, Tech. Rep., 2025.
- [4] Z. Alkhalil, C. Hewage, L. Nawaf, and I. Khan, “Phishing attacks: A recent comprehensive study and a new anatomy,” *Frontiers in Computer Science*, vol. 3, 2021.
- [5] R. M. Mohammad, F. Thabtah, and L. McCluskey, “An assessment of features related to phishing websites using an automated technique,” in *International Conference for Internet Technology and Secured Transactions*. IEEE, 2012, pp. 492–497.

- [6] A. Hannousse and S. Yahiouche, "Towards benchmark datasets for machine learning based website phishing detection: An experimental study," *Engineering Applications of Artificial Intelligence*, vol. 104, p. 104347, 2021.
- [7] A. Safi and S. Singh, "A systematic literature review on phishing website detection techniques," *Journal of King Saud University-Computer and Information Sciences*, 2023.
- [8] A. Aljofey, Q. Jiang, A. Rasool, H. Chen, W. Liu, Q. Qu, and Y. Wang, "An effective detection approach for phishing websites using url and html features," *Scientific Reports*, vol. 12, 2022.
- [9] C. Opara, B. Wei, and Y. Chen, "Htmiphish: Enabling phishing web page detection by applying deep learning techniques on html analysis," in *2020 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2020, pp. 1–8.
- [10] xorlab, "Html smuggling: How malicious actors use javascript and html to fly under the radar," 2023, accessed: 2026-04-06. [Online]. Available: <https://www.xorlab.com/en/blog/html-smuggling-how-malicious-actors-use-javascript-and-html-to-fly-under-the-radar>
- [11] Bolster Research, "How massive phishing campaigns abuse decentralized web hosting to scam victims," 2023, accessed: 2026-04-06. [Online]. Available: <https://securityboulevard.com/2023/07/how-massive-phishing-campaigns-abuse-decentralized-web-hosting-to-scam-victims/>
- [12] A. Bhardwaj, *Practical Approach to Open Source Intelligence (OSINT): Volume 2*. CRC Press, 2025.
- [13] S. Marchal *et al.*, "Phishstorm: Detecting phishing with streaming analytics," *IEEE Transactions on Network and Service Management*, vol. 11, no. 4, pp. 458–471, 2014.
- [14] M. A. Adebowale, K. T. Lwin, R. V. Sanchez, and S. Ajmal, "Intelligent web-phishing detection and protection scheme using integrated features of images, frames and text," *Expert Systems with Applications*, vol. 115, pp. 300–313, 2019.
- [15] M. S. I. Mamun *et al.*, "Detecting malicious urls using lexical analysis," in *Network and System Security: 10th International Conference, NSS 2016, Taipei, Taiwan, September 28-30, 2016, Proceedings*. Springer International Publishing, 2016, p. 10.
- [16] K. L. Chiew, R. Muniandy, A. Sulaiman, S. Abdullah, and M. F. Abdullah, "Building standard offline anti-phishing dataset for benchmarking," *International Journal of Engineering & Technology*, vol. 7, no. 4.31, pp. 7–14, 2018.
- [17] O. K. Sahingoz *et al.*, "Machine learning based phishing detection from urls," *Expert Systems with Applications*, vol. 117, pp. 345–357, 2019.
- [18] F. C. Dalgic, A. S. Bozkir, and M. Aydos, "Phish-iris: A new approach for vision based brand prediction of phishing web pages via compact visual descriptors," in *2018 2nd International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*. Istanbul, Turkey: IEEE, 2018, pp. 1–6.
- [19] M. Sánchez-Paniagua, A. García-Serrano, A. Rodríguez, and E. Martínez, "Phishing websites detection using a novel multipurpose dataset and web technologies features," *Expert Systems with Applications*, vol. 207, p. 118010, 2022.
- [20] G. Vrbančič, I. Fister Jr, and V. Podgorelec, "Datasets for phishing websites detection," *Data in Brief*, vol. 33, p. 106438, 2020.
- [21] V. A. V. Adap *et al.*, "Do not feed the phish: Phishing website detection using url-based features," in *Proceedings of the 2023 5th World Symposium on Software Engineering*, 2023.
- [22] G. Apruzzese and V. S. Subrahmanian, "Mitigating adversarial gray-box attacks against phishing detectors," *IEEE Transactions on Dependable and Secure Computing*, vol. 20, no. 5, pp. 3753–3769, 2022.
- [23] B. van Dooremaal, P. Burda, L. Allodi, and N. Zannone, "Combining text and visual features to improve the identification of cloned webpages for early phishing detection," in *Proceedings of the 16th International Conference on Availability, Reliability and Security*, ser. ARES '21. ACM, 2021.
- [24] R. Liu, Y. Lin, X. Yang, S. H. Ng, D. M. Divakaran, and J. S. Dong, "Inferring phishing intention via webpage appearance and dynamics: A deep vision based approach," in *31st USENIX Security Symposium (USENIX Security 22)*, 2022, pp. 1633–1650.
- [25] F. Castaño, E. Fidalgo Fernández, R. Alaiz-Rodríguez, and E. Alegre, "Phikita: Phishing kit attacks dataset for phishing websites identification," *IEEE Access*, vol. 11, pp. 40 779–40 789, 2023.
- [26] M. A. Tamal *et al.*, "Dataset of suspicious phishing url detection," *Frontiers in Computer Science*, vol. 6, p. 1308634, 2024.
- [27] A. Prasad and S. Chandra, "Phiusiil: A diverse security profile empowered phishing url detection framework based on similarity index and incremental learning," *Computers & Security*, vol. 136, p. 103545, 2024.
- [28] F. Çolhak, M. I. Ecevit, B. E. Uçar, R. Creutzburg, and H. Dağ, "Phishing website detection through multi-model analysis of html content," in *International Conference on Theoretical and Applied Computing*. Singapore: Springer Nature Singapore, 2024, pp. 171–184.
- [29] T. Dalton, H. Gowda, G. Rao, S. Pargi, A. Hadj Khodabakhshi, J. Rombs, S. Jou, and M. Marwah, "Phreshphish: A real-world, high-quality, large-scale phishing website dataset and benchmark," *arXiv preprint arXiv:2507.10854*, 2025. [Online]. Available: <https://arxiv.org/abs/2507.10854>
- [30] M. Erfan, P. Branco, and G.-V. Jourdan, "Domains of deception: Phishing through the lens of ownership," *Computers & Security*, p. 104860, 2026.
- [31] A. H. A. Almakhamreh and A. S. Bozkir, "Crossphire: Benefiting multimodality for robust phishing web page identification," *Applied Sciences*, vol. 16, no. 2, p. 751, 2026.
- [32] E. Zhu *et al.*, "Moe/rf: a novel phishing detection model based on revised multiobjective evolution optimization algorithm and random forest," *IEEE Transactions on Network and Service Management*, vol. 19, no. 4, pp. 4461–4478, 2022.
- [33] N. Bari *et al.*, "A filter-based feature selection framework to detect phishing urls using stacking ensemble machine learning," *Computer Modeling in Engineering & Sciences*, vol. 145, no. 1, pp. 1167–1187, 2025.
- [34] B. T. Mummadi and N. Puligundla, "Detection of phishing websites using supervised learning," *International Journal of Intelligent Systems and Applications in Engineering*, vol. 10, no. 3, pp. 171–186, Oct 2022. [Online]. Available: <https://ijisae.org/index.php/IJISAE/article/view/2153>
- [35] J. Zhou *et al.*, "A novel phishing website detection model based on lightgbm and domain name features," *Symmetry*, vol. 15, no. 1, p. 180, 2023.
- [36] K. Manguli *et al.*, "Graphish: A graph-based approach for phishing detection from encrypted tls traffic," *Journal of Information Security and Applications*, vol. 94, p. 104216, 2025.
- [37] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford, "Datasheets for datasets," *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021. [Online]. Available: <https://arxiv.org/abs/1803.09010>
- [38] A. Kulkarni, S. M. Patel, S. P. Tirmare, V. Balachandran, and T. Das, "Bridging the gap in phishing detection: A comprehensive phishing dataset collector," *arXiv preprint arXiv:2509.09592*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.09592>
- [39] OpenPhish, "Phishing feeds," 2026, official feed description page, accessed: 2026-06-17. [Online]. Available: https://openphish.com/phishing_feeds.html
- [40] Cybersecurity Presidency of the Republic of Türkiye, "Malicious links list," 2026, official malicious-link list, accessed: 2026-06-17. [Online]. Available: <https://siberguvenlik.gov.tr/zararli-baglantilar>
- [41] chromedp Project, "chromedp: A go driver for browsers supporting the chrome devtools protocol," 2026, official project repository, accessed: 2026-08-16. [Online]. Available: <https://github.com/chromedp/chromedp>
- [42] M. Gieben and Contributors, "Dns library in go," 2026, official project repository, accessed: 2026-08-16. [Online]. Available: <https://github.com/miekg/dns>
- [43] S. Hollenbeck and A. Newton, "RFC 9082: Registration data access protocol query format," RFC Editor, RFC 9082, 2021. [Online]. Available: <https://www.rfc-editor.org/info/rfc9082>
- [44] E. Foudil and Y. Shafranovich, "RFC 9116: A file format to aid in security vulnerability disclosure," RFC Editor, RFC 9116, 2022. [Online]. Available: <https://www.rfc-editor.org/info/rfc9116>
- [45] IP-API, "Ip geolocation json endpoint documentation," 2026, official documentation, accessed: 2026-08-16. [Online]. Available: <https://ip-api.com/docs/api:json>
- [46] K. Li, "Whois: A go client for domain and ip whois queries," 2026, official project repository, accessed: 2026-08-16. [Online]. Available: <https://github.com/likexian/whois>
- [47] ProjectDiscovery, "Naabu: A fast port scanner written in go," 2026, official project repository, accessed: 2026-08-16. [Online]. Available: <https://github.com/projectdiscovery/naabu>
- [48] OpenPageRank, "Top 10 million domains," 2026, domain ranking derived from the Common Crawl web graph, accessed: 2026-08-16.

- [Online]. Available: <https://openpagerank.keywordseverywhere.com/top-10-million-domains>
- [49] Google Cloud, “Web risk documentation,” 2026, official documentation, accessed: 2026-06-17. [Online]. Available: <https://docs.cloud.google.com/web-risk/docs>
- [50] PhiShark, “Phishark,” 2026, platform website. [Online]. Available: <https://phishark.io>
- [51] S. Saha Roy, U. Karanjit, and S. Nilizadeh, “A large-scale analysis of phishing websites hosted on free web hosting domains,” *arXiv preprint arXiv:2212.02563*, 2022. [Online]. Available: <https://arxiv.org/abs/2212.02563>
- [52] PhiShark2026, “Phishark2026 dataset portal,” 2026, project portal for dataset access, documentation, and dataset map. [Online]. Available: <https://phishark2026dataset.com>
- [53] H. Bijmans, T. Booi, A. Schwedersky, A. Nedgabat, and R. van Wegberg, “Catching phishers by their bait: Investigating the dutch phishing landscape through phishing kit detection,” in *30th USENIX Security Symposium (USENIX Security 21)*. USENIX Association, 2021, pp. 3757–3774. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity21/presentation/bijmans>
- [54] D. Kim, H. Cho, Y. Kwon, A. Doupe, S. Son, G.-J. Ahn, and T. Dumitras, “Security analysis on practices of certificate authorities in the https phishing ecosystem,” in *Proceedings of the 2021 ACM Asia Conference on Computer and Communications Security*, 2021, pp. 407–420. [Online]. Available: <https://dl.acm.org/doi/10.1145/3433210.3453100>
- [55] A. P. Pelekoudas, E. Bolis, J. Lindner, P. Kyriakidis, M. Davidsen, J. T. E. Hansen, C. H. Reichkender, and S. Homayoun, “Tls certificate and domain feature analysis of phishing domains in the danish .dk namespace,” *arXiv preprint arXiv:2603.21652*, 2026. [Online]. Available: <https://arxiv.org/abs/2603.21652>
- [56] OWASP Foundation, “Http security response headers cheat sheet,” 2026, official OWASP Cheat Sheet, accessed: 2026-07-10. [Online]. Available: https://cheatsheetseries.owasp.org/cheatsheets/HTTP_Header_s_Cheat_Sheet.html
- [57] F. Ji, K. Lee, H. Koo, W. You, E. Choo, H. Kim, and D. Kim, “Evaluating the effectiveness and robustness of visual similarity-based phishing detection models,” *arXiv preprint arXiv:2405.19598*, 2025.